

Shaarav S

Backend Developer · CS Undergraduate · India

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ABOUT

CS undergrad focused on backend development. I build systems around authentication, Redis caching, job queues, and REST API design — going beyond the API surface to understand the trade-offs behind each decision. I think in terms of data flow, bottlenecks, and failure modes under load. I also maintain a fully customized Arch-based Linux environment, tuning schedulers and system-level settings for low-latency workloads — the same systems thinking applied to infrastructure. Quick to pick up new tools, self-sufficient when researching, and looking for backend roles where I can take real ownership.

TECHNICAL SKILLS

Languages	JavaScript (Node.js), Python, SQL, Bash / Fish
Backend	Express.js, REST API Design, JWT Auth + Refresh Tokens, bcrypt, BullMQ, Rate Limiting
Databases	PostgreSQL, Redis (caching, queues, pub/sub)
Frontend	React, Vite, Tailwind CSS
Tools	Git, GitHub, Docker, Postman, npm, SSH
Environment	Linux (CachyOS / Arch), Fish shell, CLI-native workflow, system configuration
Interests	Network Security, AppSec, DevSecOps, System Design, OS Internals

PROJECTS

URL Shortener — *Express.js · PostgreSQL · Redis · JWT · bcrypt*

- Built a complete URL shortening service with user authentication (JWT + bcrypt), per-user link management, and redirect resolution.
 - Implemented Base62 encoding from scratch — understood the math and wrote it without a library.
 - Redis caching layer for redirect lookups, reducing repeated database reads on high-traffic links.
 - JWT middleware for protected routes; cleanly handled token expiry, invalid states, and error responses.
- github.com/saltyip/urlshortnercongential

Email Job Queue System — *BullMQ · Redis · PostgreSQL · Node.js · Gmail SMTP · Railway*

- Email job queue with BullMQ — producers enqueue jobs, workers process and dispatch via Gmail SMTP.
 - PostgreSQL job logging for a persistent audit trail: job status, timestamps, and failure reasons survive restarts.
 - Exponential backoff retry logic for transient SMTP failures and dead-letter scenarios.
 - Real-time status endpoint to query job state; deployed and running on Railway.
- github.com/saltyip/emailqueue

Dev Tooling & Shell Scripts — *Fish Shell · Bash · GNOME · GNU Stow · CachyOS*

- Fish shell scripts to automate repetitive dev tasks and streamline personal workflow.
 - dmenu-based launchers with custom keybindings on a configured GNOME/tiling setup.
 - Fully customized Arch Linux system: boot optimization, CPU scheduler tuning (scx_lavd), SSH hardening.
- github.com/saltyip/saltydotfiles

SIDE PROJECTS

Conversation Simulator · *Node.js · Ollama (self-hosted LLM)*

Self-hosted AI free conversation simulation with an imposter-style minigame also — one AI running multiple personality prompts running locally via Ollama, hidden among human-style responses. Built with Node.js.

EDUCATION

B.Tech — Computer Science 2024 – 2028 India · 2nd Year

Relevant Coursework: Design & Analysis of Algorithms, Probability & Statistics, Operating Systems, Computer Networks

ADDITIONAL

- Pursuing cybersecurity fundamentals through CTF competitions (PicoCTF) and platforms including TryHackMe and HackTheBox Starting Point.
- Linux system optimization: CPU scheduler tuning (scx_lavd), boot time reduction, and latency optimization on an Arch-based system.
- Personal portfolio at shaarav.dev featuring a custom interactive OS-style interface built from scratch.